Mahidol University
Faculty of Information
and Communication Technology

Enhanced Adversarial Drift Detection for MLOps Feature Monitoring

The 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE 2026) Bangkok, Thailand

Nusrat Begum
Faculty of ICT
Mahidol University

Phaphontee Yamchote
Faculty of ICT
Mahidol University

Chainarong Amornbunchornvej
NECTEC, NSTDA

Thanapon Noraset
Faculty of ICT
Mahidol University

This research project was partially supported by Faculty of Information and Communication Technology, Mahidol University.

Table of Contents

01 Drift & Feature Drift

02 Motivation & Problem

03 Contributions

04 EADD Pipeline

05 Experimental Setup

06 Results

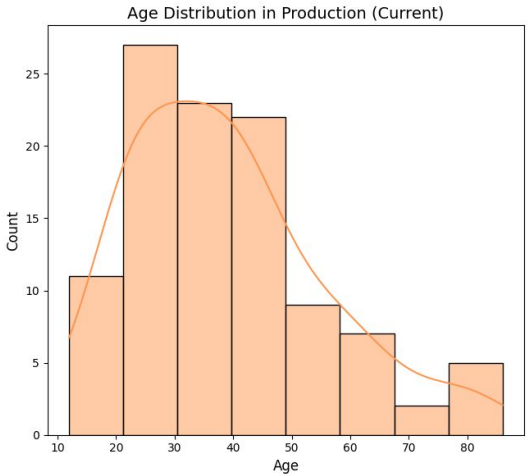
07 Limitations & Future Work

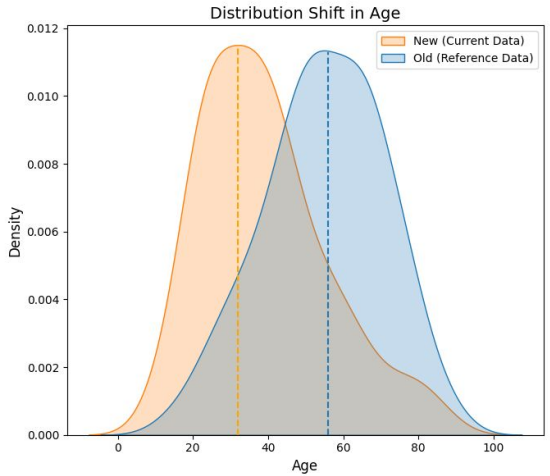
08 Conclusion

Drift & Feature Drift

Definition: X = Feature (Age) Changed [1]







[1] Concept drift: Lu et al., 2019; Gama et al., 2014

Motivation: Unsupervised Drift Detection (D3)

Core Concept: Adversarial Validation. Formulated in competitive data science to detect dataset shift between training and testing distributions by training a binary discriminator (Fast.ai / Kaggle consensus, c. 2016).

The Stream Algorithm (D3) [2][3][4]

Concept: Lightweight framework for continuous data streams.

Mechanism:

- Splits streams into sliding windows of "old" vs "new"
- Trains a classifier to separate windows
- Signals drift if AUC exceeds threshold (τ)

Primary Use: Real-time stream monitoring.

Enterprise Usage (Uber) [14]

Concept: Adversarial validation for batch marketing.

Mechanism:

- Adversary model checks new batch vs training data
- Adversarial Feature Selection identifies drift source
- Drops drifted features instead of full retraining

Primary Use: Fixing batch concept drift at scale.

[2] D3: Gözüaçık et al., 2019
[3] Benchmark: Lukats et al., 2024
[4] C2ST: Lopez-Paz & Oquab, 2017
[14] Adversarial validation (Uber): Pan et al., 2020

Problem and Research Gap

Problem Statement

Distribution drift degrades models in production. Existing detectors only answer:

Was it drift? YES/NO (Binary alarm)

Practitioners need to know:

- Which features drifted?
- How severe is the shift?
- Is it real or autocorrelation noise?

Current Limitations

One of the most robust unsupervised detectors is D3 [3]

Identified Gaps:

- No formal statistical test for significance
- No feature attribution
- Linear classifier misses non-linear drift

[3] Benchmark: Lukats et al., 2024

Main Contributions

1. ASPT [5]

Adaptive Sequential Permutation Test

- Shuffle the data
- **90–95%** fewer permutations
- Formal permutation test gates drift

2. Localisation [6]

SHAP-Based Localisation + DSI

- **SHAP:** Identifies feature drift
- **DSI:** Quantifies drift severity

3. Benchmark [3]

Experimental Setup (3 Datasets)

1. Synthetic
2. Real World
3. SHAP Test Data

Key Study Findings

EADD vs. 7 latest unsupervised detectors

[5] ASPT: Gandy, 2009
[6] SHAP: Lundberg & Lee, 2017
[3] Benchmark: Lukats et al., 2024

EADD Pipeline [7][8][4]

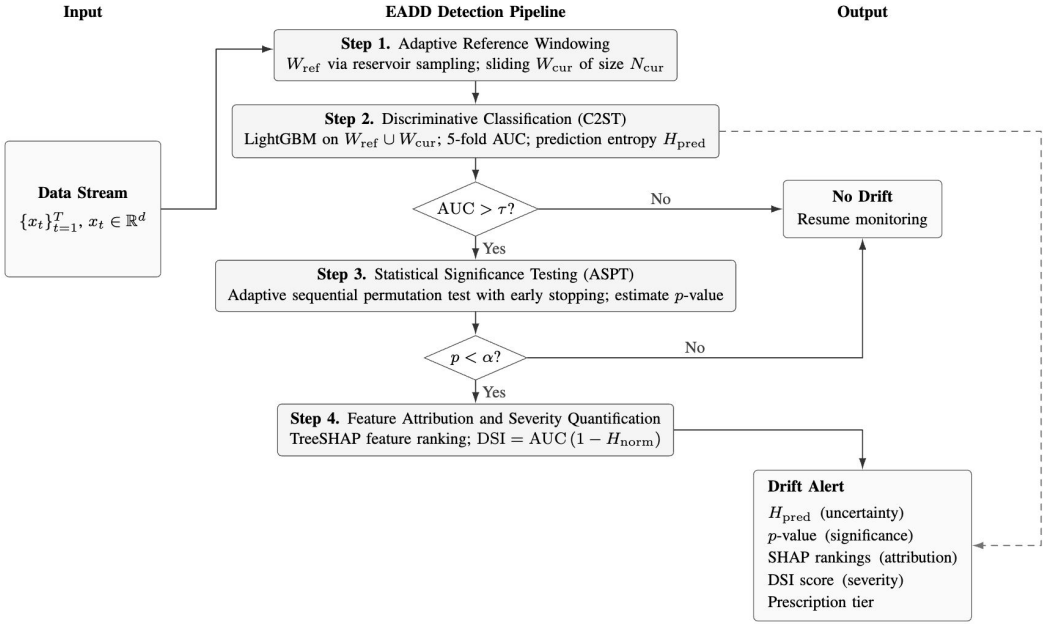


Fig. 1. EADD detection pipeline. W_{cur} is compared against W_{ref} via a discriminative classifier; H_{pred} provides a continuous uncertainty monitoring signal (dashed). AUC $> \tau$ triggers ASPT for statistical confirmation (Step 3); upon significance, TreeSHAP and DSI localise and quantify the shift (Step 4).

[7] Reservoir sampling: Vitter, 1985
[8] LightGBM: Ke et al., 2017
[4] C2ST: Lopez-Paz & Oquab, 2017

Contribution 1: ASPT

Problem

D3 uses fixed AUC threshold → false alarms on autocorrelated data. [2]

Solution

Adaptive Early Stopping (ASPT) [5]

ASPT Decision Rule (i = shuffle/permute)

Early Reject H_0 (Drift Confirmed)

If $c_i = 0$, $i \geq B_{min}$, $1/(i+1) < \alpha \rightarrow p = 1/(i+1)$

Early Accept H_0 (No Drift)

If $c_i/i > 2\alpha$, $i \geq B_{min} \rightarrow p = (c_i+1)/(i+1)$

Terminal State

$p = (c_B_{max} + 1) / (B_{max} + 1)$

Efficiency Result

90-95%

Permutation Savings

- Drift: i = 20 (Bmax=199)
- Overall α preserved

[5] Permutation test: Gandy, 2009
[2] D3: Gözüaçık et al., 2019

Contribution 2: SHAP Attribution + DSI

SHAP on the Discriminative Classifier [6][9]

TreeSHAP on the LightGBM gate (which D3 discards):

$$\bar{\phi}_j = (1/n) \Sigma |\phi_{ij}|, s_j = \bar{\phi}_j / \Sigma \bar{\phi}_k$$

⇒ Ranks features by drift contribution.

Drift Severity Index (DSI) (Shannon, 1948)

$$DSI = AUC \times (1 - H_{norm}(s))$$

$H_{norm} \in [0, 1]$ via $\log_2 d$ normalisation → dimensionality-invariant.

SHAP + DSI

Combines drift detection (AUC) with feature importance (SHAP) for a robust metric.

[6] SHAP: Lundberg & Lee, 2017
[9] TreeSHAP: Lundberg et al., 2020
[10] Entropy: Shannon, 1948

Contribution 3: Experimental Setup

DATASETS

Synthetic: (n=5000, d=5) 4 drift patterns + 4 stable streams.

Real-world: 13 datasets [11], 4 annotated INSECTS + 9 unannotated [3].

SHAP Data: (n=10,000, d=10, t=5k)

EVALUATION

SHAP:
(n=10,000, d=10, t=5000) tested on univariate / subset analysis.

Benchmark:
D3 vs 7 Unsupervised Models (BNDM, CSDDM, IBDD, OCDD, SPLL, UDetect).

LIGHTGBM MODEL CONFIGURATION PARAMETERS

Model: **LightGBM (50 trees)**

Sliding Window: **Nref = 500, Ncur = 200, Step(M) = 50**

ASPT Tuning: **Bmax = 199, Bmin = 10, α = 0.05, τ = 0.7**

n = sample size
d = features/columns
t = drift start point
Nref = Reference Window
Ncur = Current Window
M = Steps
Bmax = Max Shuffle
Bmin = Min Shuffle
α = Significance
τ = AUC

[11] INSECTS: Souza et al., 2020
[3] Benchmark protocol: Lukats et al., 2024

Result 1: Synthetic Coverage + False Alarms

Key Finding: EADD is the only detector achieving full coverage with 0 false alarms.

Analysis of Performance [3]

Mean False Alarm (Mean FA):

EADD, D3, and SPLL achieve a flawless 0.00.

Winning Row (EADD):

- 100% Coverage (4/4 Types)
- Zero Mistakes (0.00 Mean FA)

Detector	Types	Coverage	Mean FA
EADD	**4/4**	**100%**	**0.00**
BNDM	4/4	100%	4.75
CSDDM	4/4	100%	5.50
IBDD	4/4	100%	46.25
OCDD	4/4	100%	17.50
D3	2/4	50%	0.00
SPLL	2/4	50%	0.00
UDetect	0/4	0%	n/a

[3] Baselines & reporting: Lukats et al., 2024

Result 2: Real-World INSECTS — Faster Detection [11][3]

Dataset	Prec.	Recall	MTD (EADD)	MTD (D3)
InsectsAbrupt	0.33	0.60	**135**	155.5
InsectsGradual	n/a	0/1	n/a	n/a
InsectsIncrAbrupt	0.14	1.00	**33**	272.5
InsectsIncrRecur	0.17	1.00	**133**	323.5

FASTER DETECTION

EADD outperforms D3 across the board, showing superior speed in drift identification.

MASSIVE GAP

On **InsectsIncrAbrupt**, EADD is **~8x faster** (33 vs 272.5).

GRADUAL DRIFT

Neither detector found the drift in **InsectsGradual** (Recall 0/1), highlighting a shared limitation.

[11] INSECTS: Souza et al., 2020
[3] Lukats et al., 2024

Result 3: SHAP Attribution Accuracy [6][9]

Fidelity: P@k = R@k = 1.0 (Precision and Recall) on univariate SHAP attribution.

Scenario	AUC	P@k	R@k	Top feature (%)	DSI
Univariate (1/10)	0.784	1.0	1.0	F3 (52.3%)	0.193
Subset (3/10)	0.809	1.0	1.0	F5 (24.5%)	0.096

Analysis of Performance

Meaningful Result: univariate (1/10 features correctly ranked first).

DSI Order

Orders by concentration: **univariate > subset**.

[6] SHAP: Lundberg & Lee, 2017
[9] TreeSHAP: Lundberg et al., 2020

Limitations & Future Work

Limitations

Wall-clock: ~50–160 × D3 (dominated by LightGBM training every M=50 steps).

Multiple testing: ~600 tests/30K samples; α′ ≈8.3×10−5 floor. Zero-FA result is empirical.

DSI boundaries: Heuristics, never triggered (all DSI < 0.3).

Attributions: SHAP is correlational, not causal.

Sampling: No isolation of reservoir sampling vs. window config contributions.

Future Work

- **Causal attribution** methods [12].
- Anytime-valid **e-values** for streaming FWER control [13].
- Multi-domain **DSI calibration** to ground tier boundaries.
- **Adversarial robustness** of discriminative detectors.

[12] Causal attribution: Budhathoki et al., 2021
[13] e-values: Pandeva et al., 2024

Conclusion

Summary: EADD provides a unified pipeline to answer whether, when, which, and how severe drift is in data streams.

Technical Extensions

1. ASPT + Hpred: Rigorous confirmation with 90–95% perm savings.

2. SHAP + DSI: Feature-level diagnosis and severity quantification.

3. Benchmarking: Validated vs. 7 unsupervised detectors on diverse datasets.

Headline Results

Robustness: 0 false alarms vs D3 on autocorrelated streams (D3: 87.4 at τ=0.6, Mann–Whitney p=0.0101).

Coverage: 4/4 synthetic — only detector to achieve both.

Speed: Up to 8× faster MTD than D3 on INSECTS.

Fidelity: P@k = R@k = 1.0 on univariate SHAP attribution.

Thank you!

Selected References (Full list in paper & GitHub)

1. J. Lu, et al., "Learning under concept drift: A review," 2019 / J. Gama, et al., "A survey on concept drift adaptation," 2014.
2. Ö. Gözüaçık, et al., "Unsupervised concept drift detection with a discriminative classifier (D3)," 2019.
3. D. Lukats, et al., "A benchmark and survey of fully unsupervised concept drift detectors on real-world data streams," 2024.
4. D. Lopez-Paz, et al., "Revisiting classifier two-sample tests (C2ST)," 2017.
5. A. Gandy, "Sequential implementation of Monte Carlo tests with uniformly bounded resampling risk," 2009.
6. S. M. Lundberg, et al., "A unified approach to interpreting model predictions (SHAP)," 2017.
7. J. S. Vitter, "Random sampling with a reservoir," 1985.
8. G. Ke, et al., "LightGBM: A highly efficient gradient boosting decision tree," 2017.
9. S. M. Lundberg, et al., "From local explanations to global understanding with explainable AI for trees (TreeSHAP)," 2020.
10. C. E. Shannon, "A mathematical theory of communication," 1948.
11. V. M. A. Souza, et al., "Challenges in benchmarking stream learning algorithms with real-world data (INSECTS)," 2020.
12. K. Budhathoki, et al., "Why did the distribution change?," 2021.
13. T. Pandeva, et al., "E-valuating classifier two-sample tests," 2024 (Optional).
14. J. Pan, et al., "Adversarial validation approach to concept drift at Uber," 2020.